

Discrete Mathematics

Week 13: Graphs Part 2

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Directed Graph Terminology

Definition 4: When (u, v) is an edge of the graph G with directed edges, u is said to be **adjacent to** v and v is said to be **adjacent from** u .

- The vertex u is called the **initial vertex** of (u, v) , and v is called the **terminal** or **end vertex** of (u, v) .
- The initial vertex and terminal vertex of a loop are the same.

Directed Graph Terminology

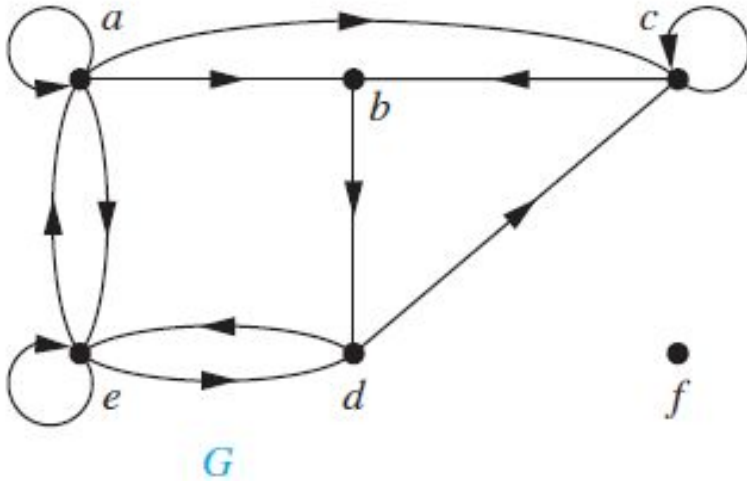
Definition 5: In a graph with directed edges the **in-degree of a vertex v** , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex.

The **out-degree of v** , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.

- Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.

Directed Graph Example

Example: Find the in-degree and out-degree of each vertex in the graph G with directed edges.



Directed Graph

Theorem 3: Let $G = (V, E)$ be a graph with directed edges. Then

$$|E| = \sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v).$$

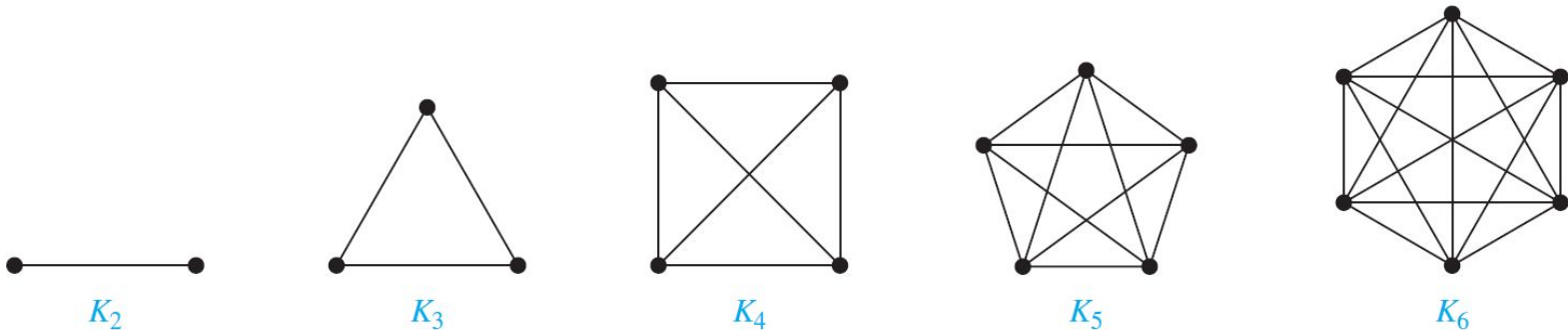
Proof:

- Because each edge has an initial vertex and a terminal vertex, the sum of the in-degrees and the sum of the out-degrees of all vertices in a graph with directed edges are the same.

3. Special Types of Simple Graphs

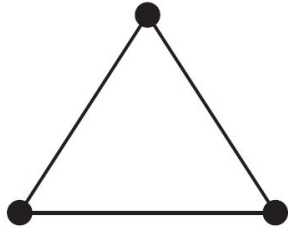
A **complete graph** on n vertices, denoted by K_n , is a simple graph that contains exactly one edge between each pair of distinct vertices.

A simple graph for which there is at least one pair of distinct vertex not connected by an edge is called **noncomplete**.

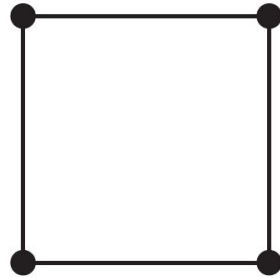


Special Types of Simple Graphs

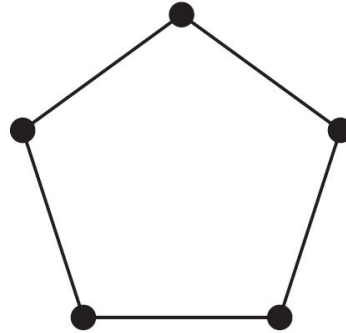
A **cycle** C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.



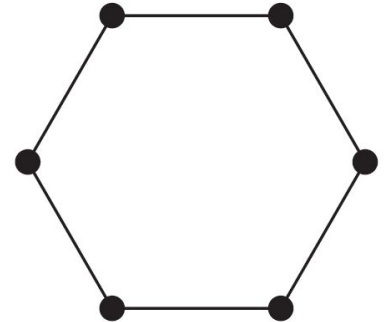
C_3



C_4



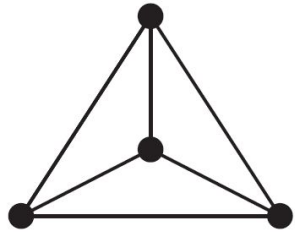
C_5



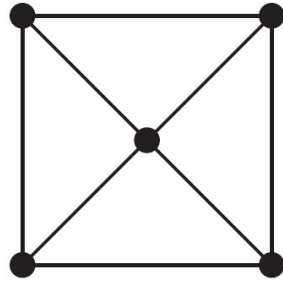
C_6

Special Types of Simple Graphs

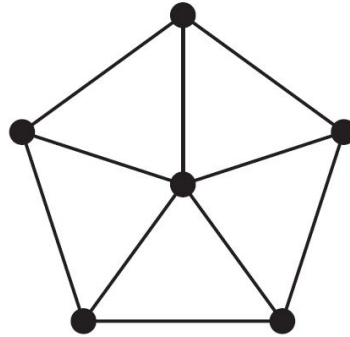
A **wheel** W_n is obtained by adding an additional vertex to a cycle C_n , for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n by new edges.



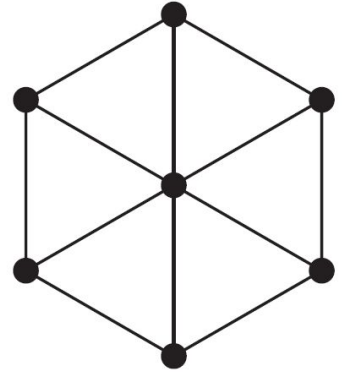
W_3



W_4



W_5



W_6

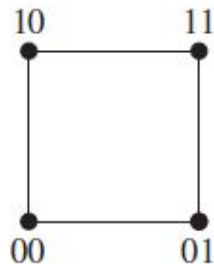
Special Types of Simple Graphs

An **n-dimensional hypercube**, or **n-cube**, denoted by Q_n , is a graph that has vertices representing the 2^n bit strings of length n .

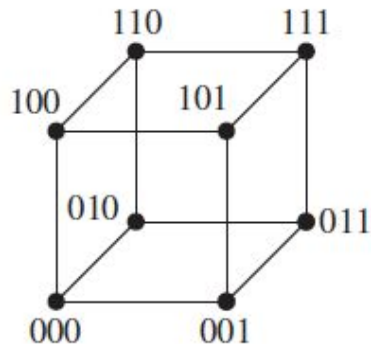
Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.



Q_1



Q_2



Q_3

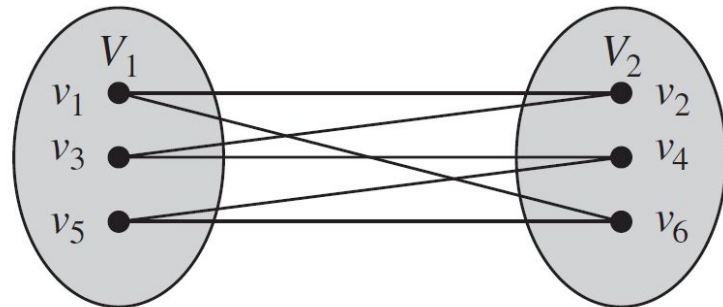
Bipartite Graphs

Definition 6: A simple graph is called **bipartite** if its vertex set V can be partitioned into 2 disjoint subsets V_1 and V_2 such that each edge connects a vertex from one subset to a vertex of the other.

- No edge connects either two vertices in V_1 or two vertices in V_2

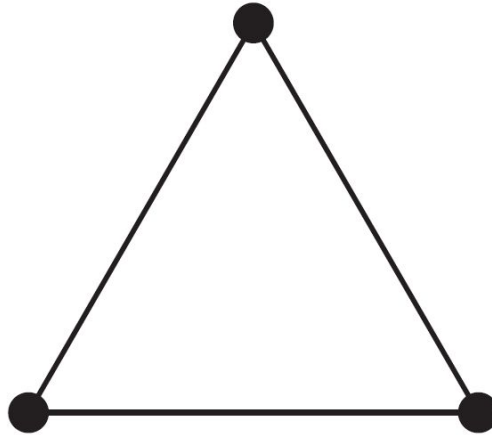


C_6



Example

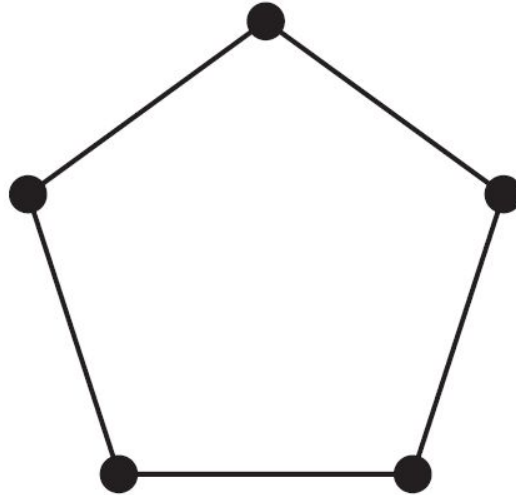
Example 5: Is this graph bipartite?



K_3

Example

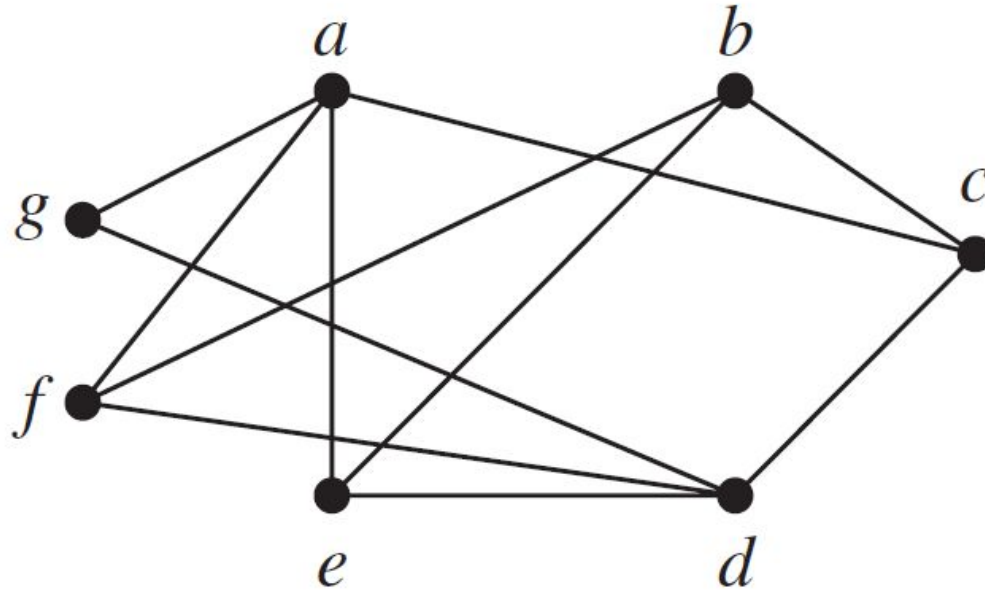
Example 6: Is this graph bipartite?



C_5

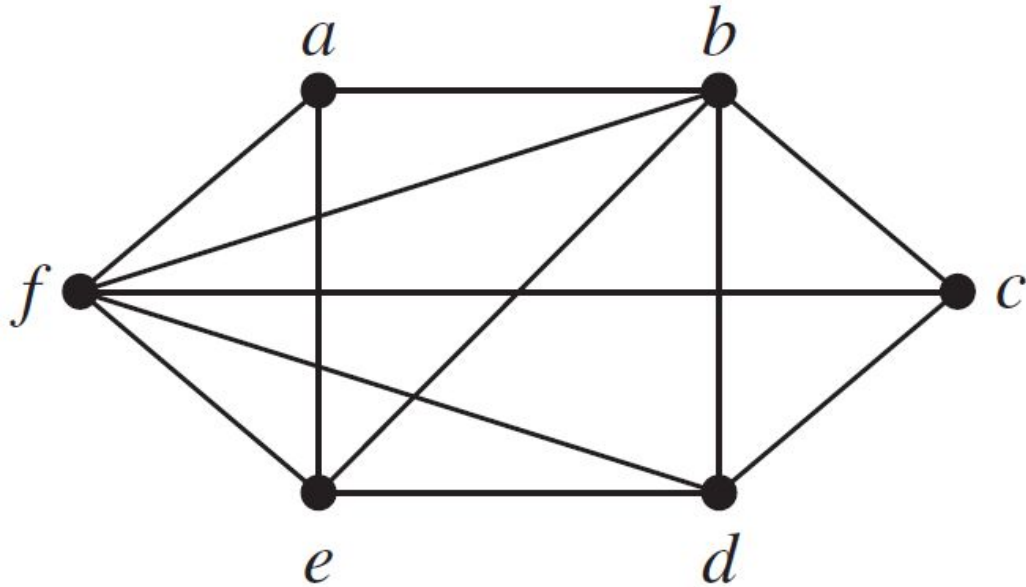
Exercise

Exercise 5: Is this graph bipartite?



Exercise

Exercise 6: Is this graph bipartite?

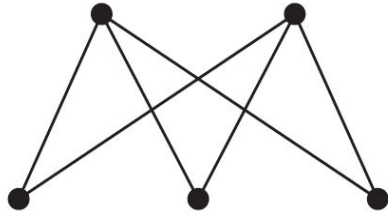


Complete Bipartite Graphs

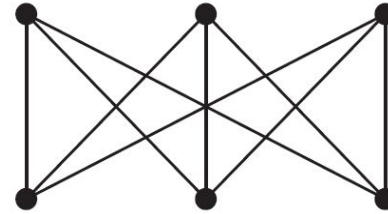
Definition 7: A **complete bipartite graph** $K_{m,n}$ is a bipartite graph with subsets of m and n vertices, respectively with an edge between each pair of vertices from opposite subsets.

- An edge between two vertices where one vertex is in the first subset and the other vertex is in the second subset.

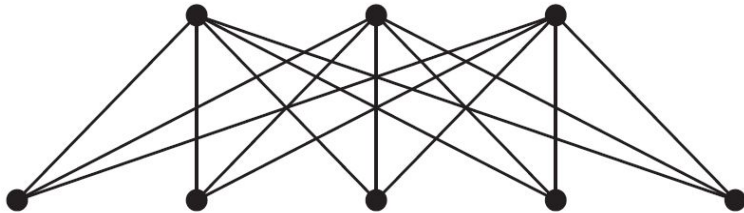
Complete Bipartite Graphs



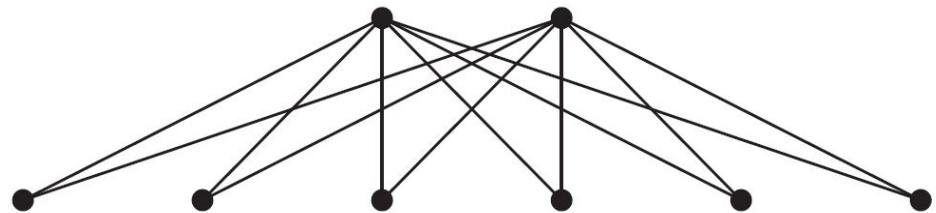
$K_{2,3}$



$K_{3,3}$



$K_{3,5}$



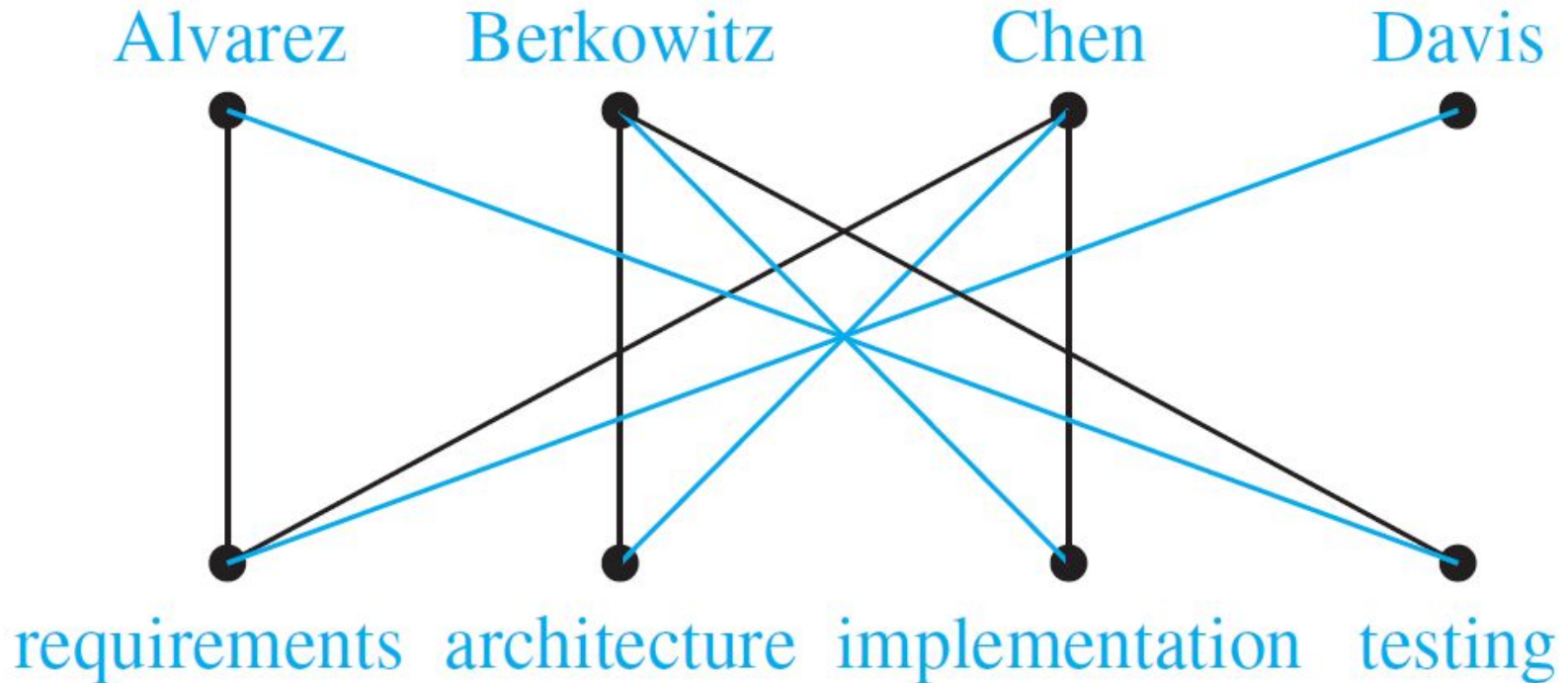
$K_{2,6}$

Bipartite Graphs and Matchings

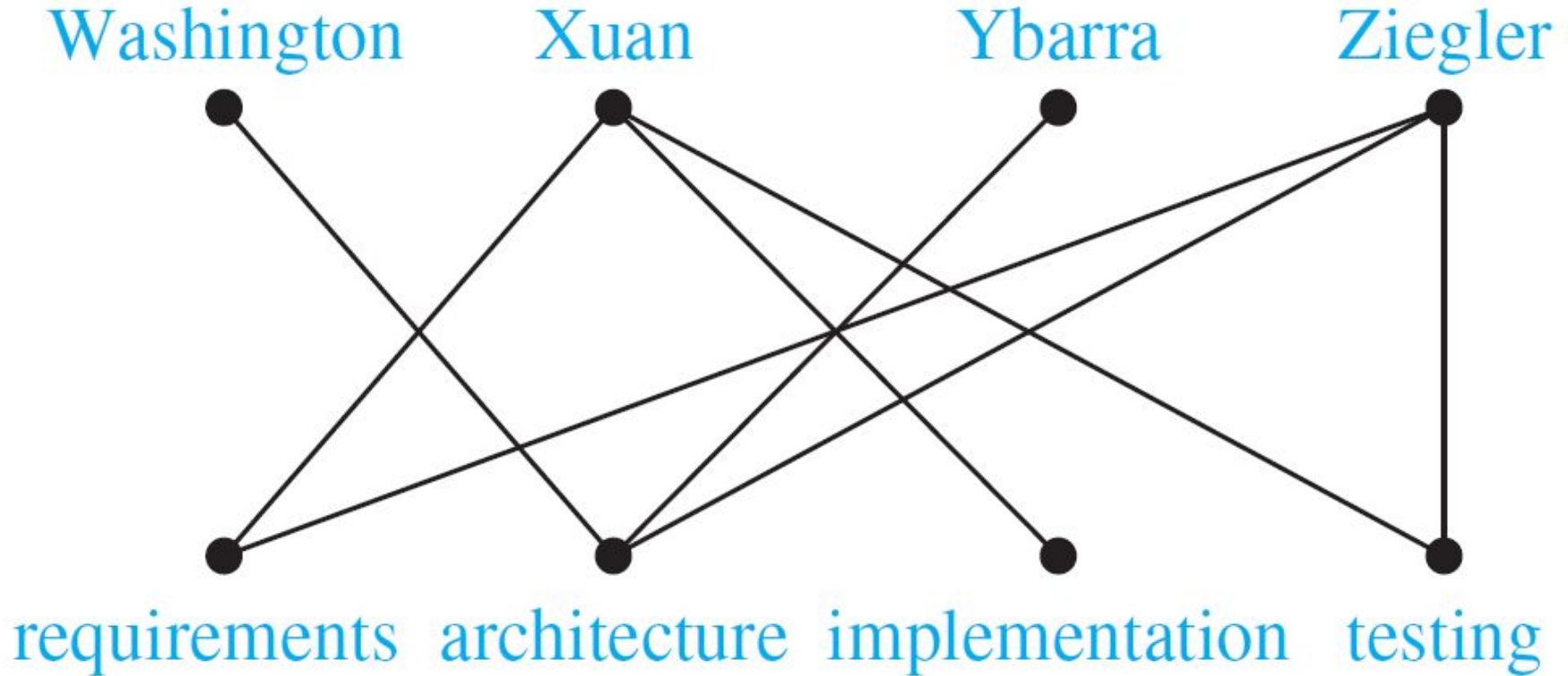
Bipartite graphs are used to model applications that involve **matching** the elements of one set to elements in another.

- **Job assignments** - Suppose that there are m employees in a group and n different jobs that need to be done, where $m \geq n$. Each employee is trained to do one or more of these n jobs.
 - **Task:** Assign an employee to each job such that no employee is assigned more than one job.

Task: Assign an employee to each job such that no employee is assigned more than one job.



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Bipartite Graphs and Matchings

Finding an assignment of jobs to employees or organizing team to roles can be thought of as finding a matching in the graph model.

Definition 8: A **matching** M in a simple graph $G = (V, E)$ is a subset of the set of edges, E , such that no two edges are incident with the same vertex.

Bipartite Graphs and Matchings

Definition 9: A **maximum matching** is a matching with the largest number of edges.

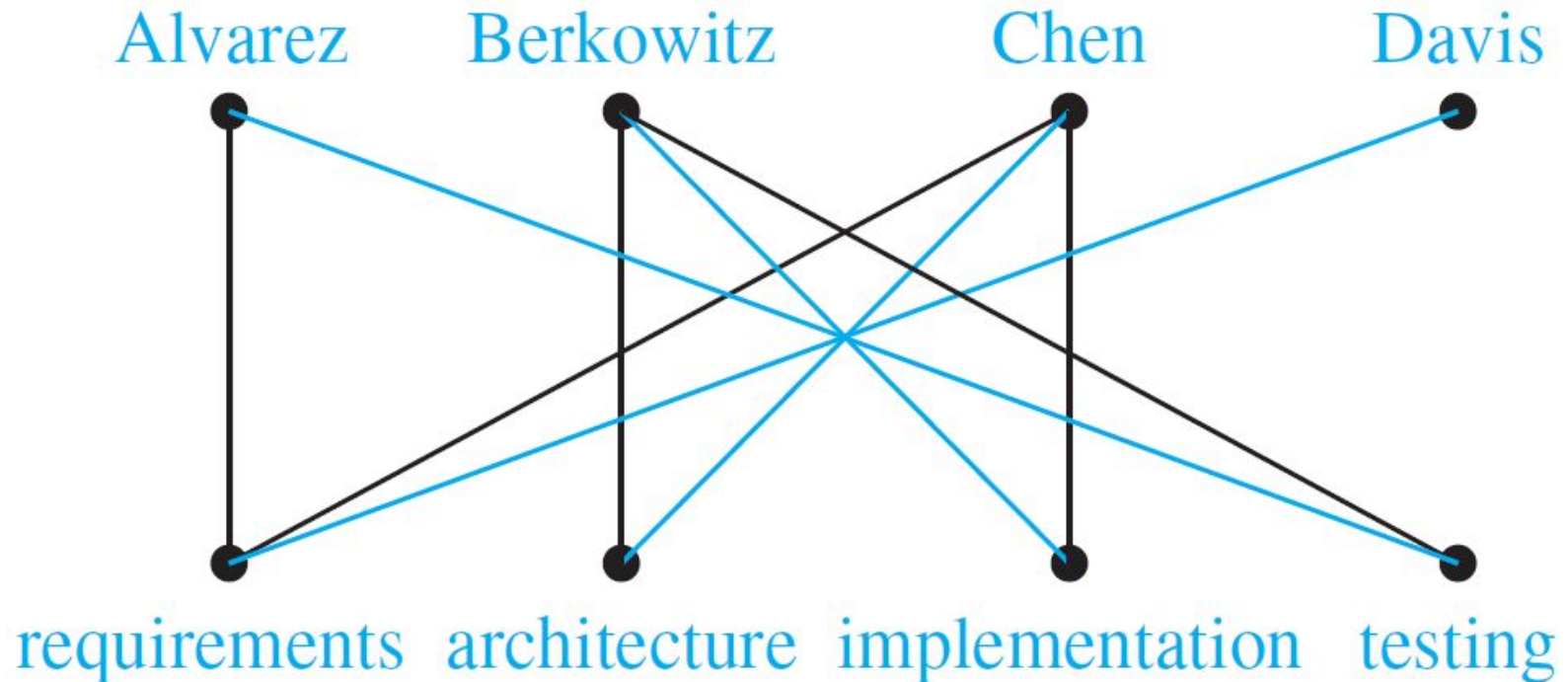
- Assign jobs to employees so that the largest number of jobs are assigned to employees.
 - Maximum matching from set of jobs to set of employees

Bipartite Graphs and Matchings

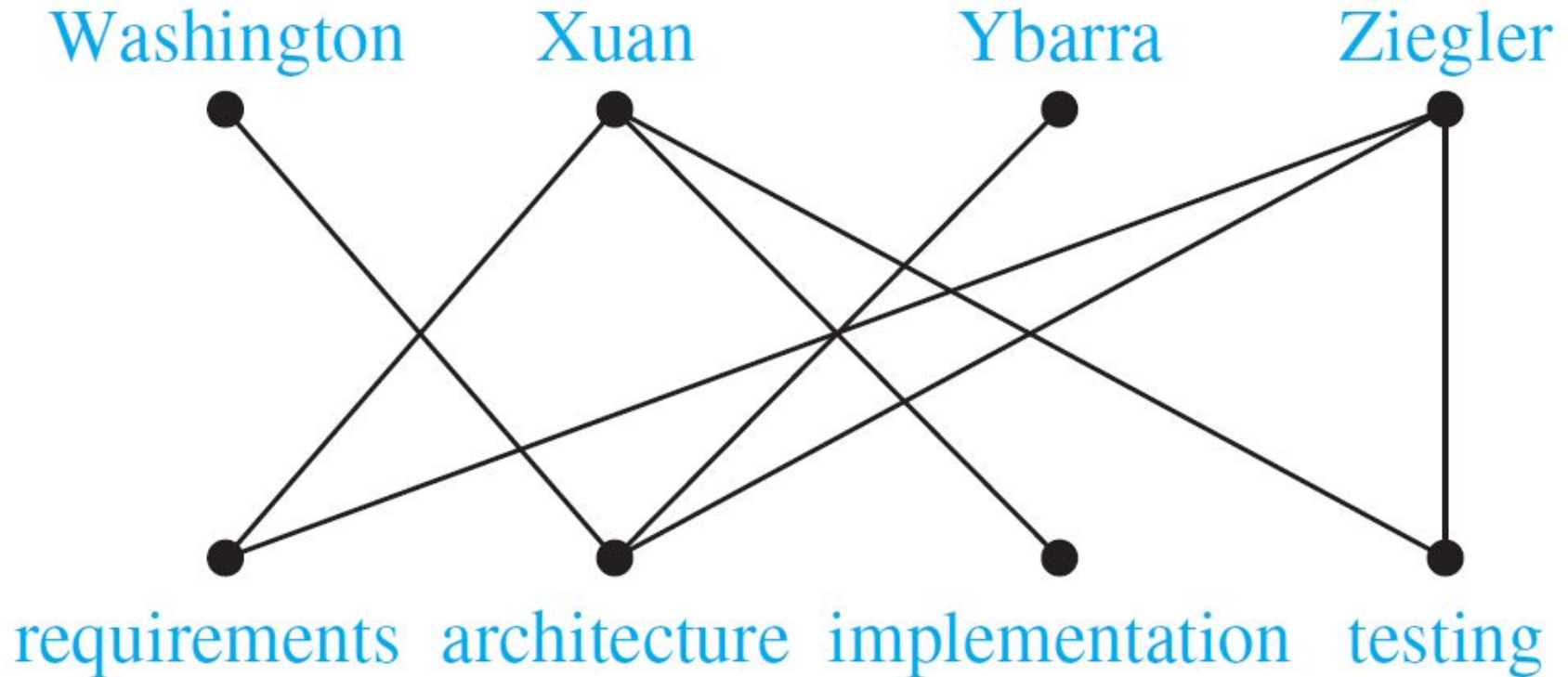
Definition 10: A matching M is a **complete matching** from V_1 to V_2 if every vertex in V_1 is the endpoint of an edge in the matching.

- Or, if $|M| = |V_1|$
- Assign employees to all jobs
 - Complete matching from set of jobs to set of employees

Task: Find a complete matching and maximum matching from the set of jobs to the set of employees



Task: Find a complete matching and maximum matching from the set of jobs to the set of employees



Marriages on an Island

Suppose that there are m men (V_1) and n women (V_2) on an island.

Each person has a list of members of the opposite gender acceptable as a spouse.

A maximum matching is a largest possible set of married couples.

A complete matching of V_1 is a set of married couples where every man is married, but possibly not all women.

Hall's Marriage Theorem

Theorem 4: The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|N(A)| \geq |A|$ for all subsets A of V_1 .

4. Connectivity

Paths

Informal Definition: A *path* is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph.

Bangkok --- Saraburi --- Korat

Applications: Numerous problems can be modeled with paths formed by traveling along edges of graphs such as:

Paths in Undirected Graph

Definition 1: Let n be a nonnegative integer and G an undirected graph.

A **path** of length n from u to v in G is a sequence of n edges $e_1, \dots, e_n$ of G for which there exists a sequence $x_0 = u, x_1, \dots, x_{n-1}, x_n = v$ of vertices such that e_i has, for $i = 1, \dots, n$, the endpoints x_{i-1} and x_i .

When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \dots, x_n$.

Paths in Undirected Graph

Definition 2:

The path is a **circuit** if it begins and ends at the same vertex, that is, if $u = v$, and has length greater than zero.

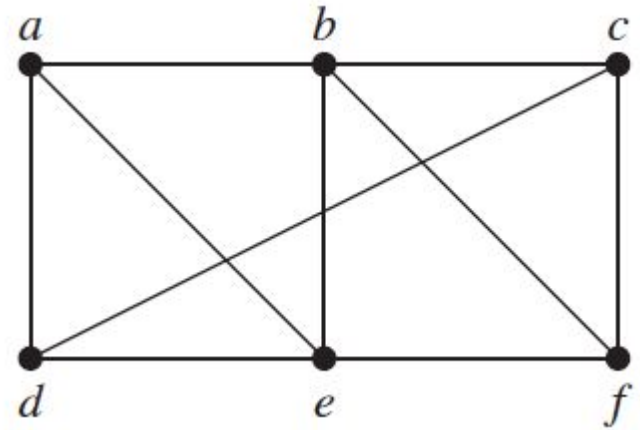
A path or circuit is **simple** if it does not contain the same edge more than once.

- The path or circuit is said to **pass through** the vertices $x_1, x_2, \dots, x_{n-1}$ or **traverse** the edges $e_1, e_2, \dots, e_n$.

Path Example

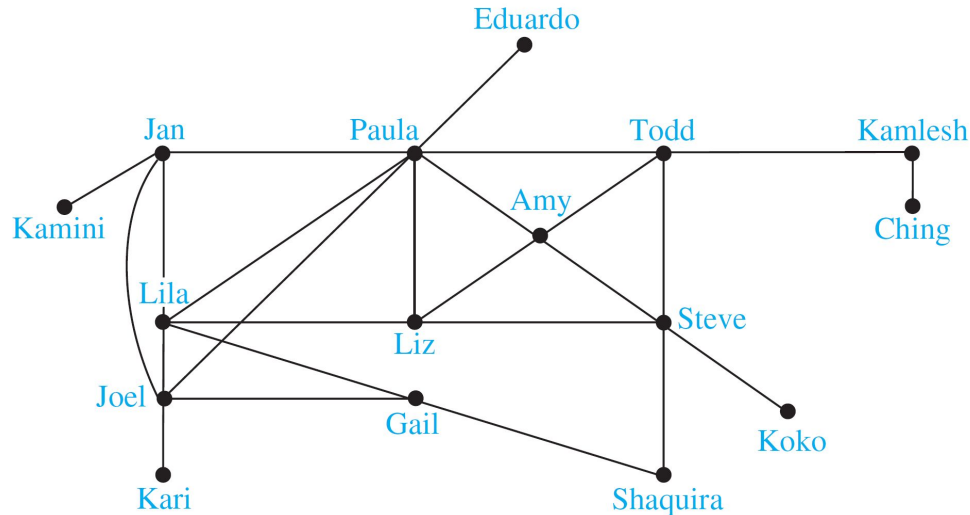
Which sequence is a path, circuit, or simple?

- a, d, c, f, e
 - Edges traversed: {a, d}, {d, c}, {c, f}, and {f, e}
- d, e, c, a
- b, c, f, e, b
- a, b, e, d, a, b



Paths in Acquaintanceship Graphs

In an acquaintanceship graph there is a path between 2 people if there is a chain of people linking these people, where 2 people adjacent in the chain know one another.



Six Degrees of Separation

- Many social scientists have speculated that almost every pair of people in the world are linked by a small chain of **no more than six**, or fewer.
- This would mean that almost every pair of vertices in the acquaintanceship graph containing all people in the world is linked by a path of length not exceeding five.

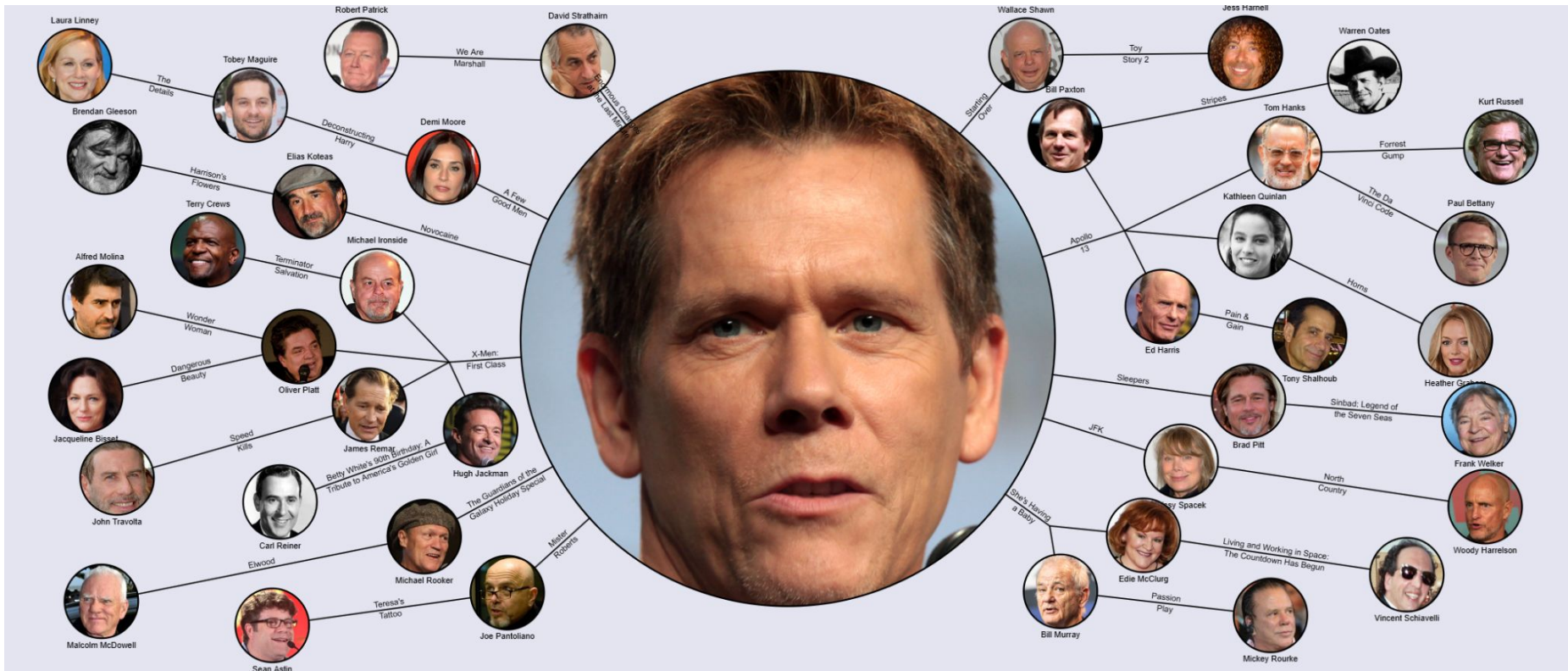
Paths in Collaboration Graphs: Bacon Number

- In the Hollywood graph, two actors a and b are linked when there is a chain of actors linking a and b , where every two actors adjacent in the chain have acted in the same movie.
- The **Bacon number** of an actor c is defined to be the length of the shortest path connecting c and the well-known actor Kevin Bacon.
- <https://oracleofbacon.org/graph.php>

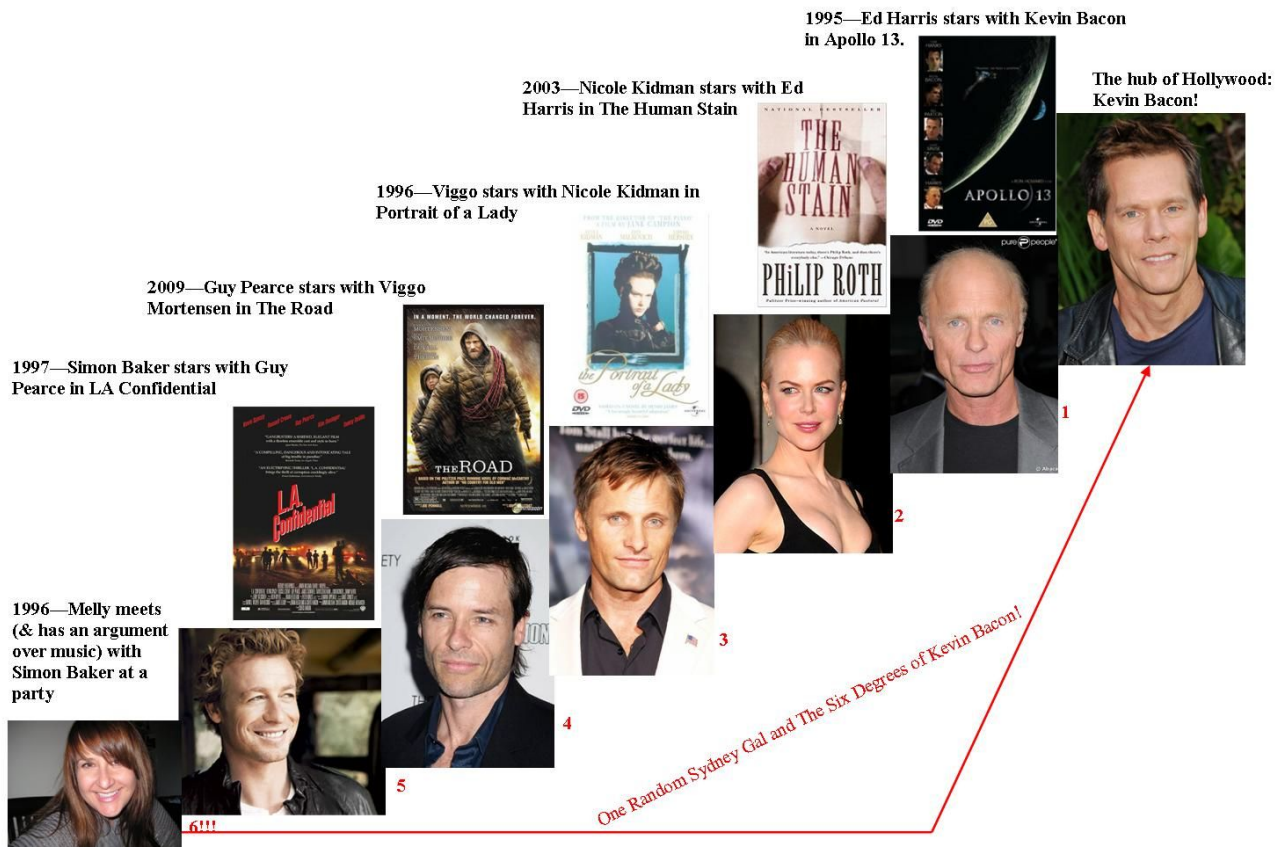
TABLE 2 The Number of Actors with a Given Bacon Number (as of August 2017).

<i>Bacon Number</i>	<i>Number of People</i>
0	1
1	3,452
2	401,636
3	1,496,104
4	390,878
5	4,388
6	631
7	131
8	9
9	1

Six Degrees of Kevin Bacon



Six Degrees of Kevin Bacon



Connectedness in Undirected Graphs

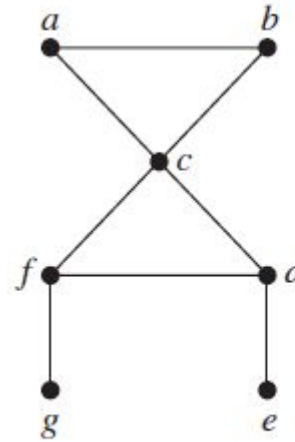
Definition 3: An undirected graph is called ***connected*** if there is a path between every pair of distinct vertices of the graph.

An undirected graph that is not connected is called ***disconnected***.

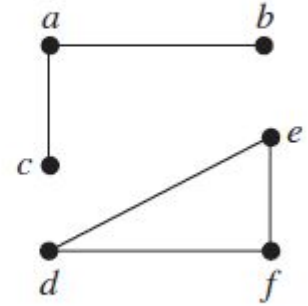
Connectedness in Undirected Graphs

We say that we **disconnect** a graph when we remove vertices or edges, or both, to produce a disconnected subgraph.

For instance, any 2 computers on a network can communicate if and only if the graph is connected.



G_1

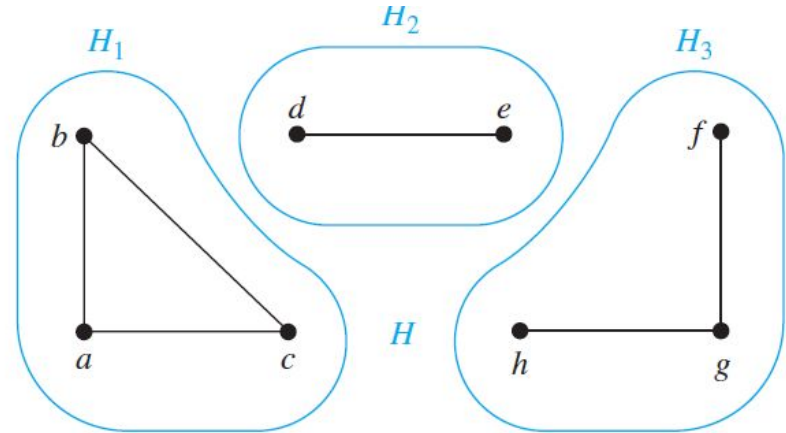


G_2

Connected Components

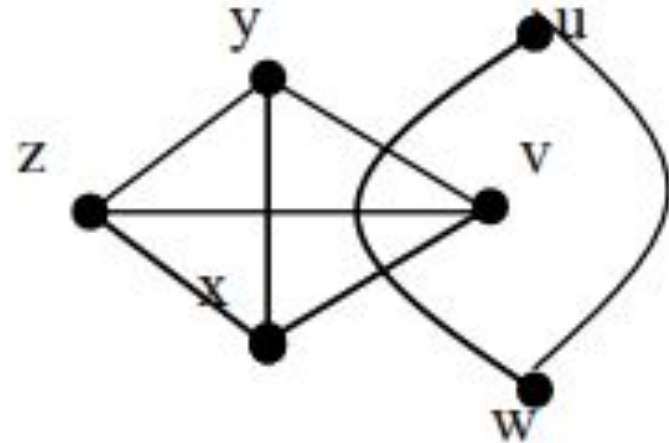
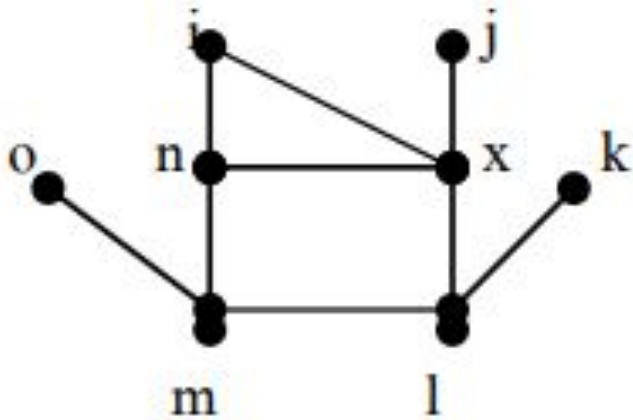
Definition 4: A **connected component** of a graph G is a connected subgraph of G that is not a proper subgraph of another connected subgraph of G .

A graph G that is not connected has two or more connected components that are disjoint and have G as their union.



Connected Graphs Example

Example: Which of the following graphs is connected or disconnected?



Cut Vertices and Cut Edges

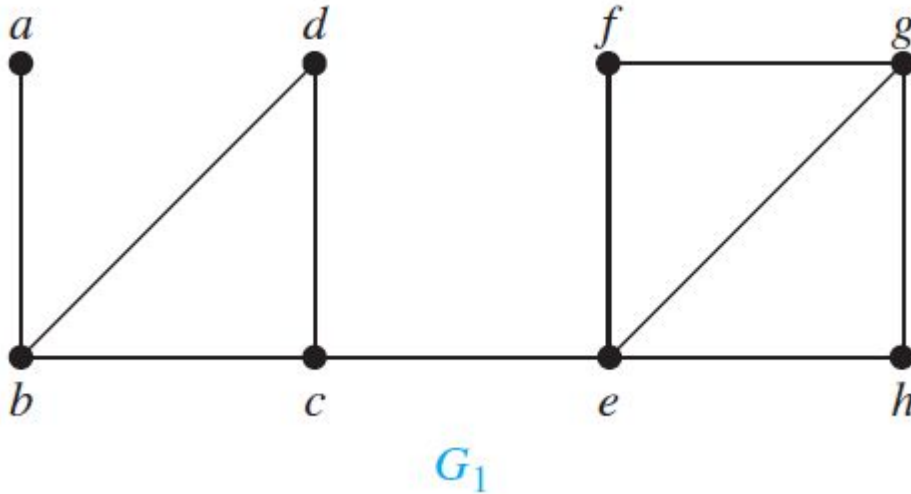
Suppose that a graph represents a computer network. A connected graph indicates that any 2 computers on the network can communicate.

Definition 5: The removal of a vertex from a graph and all incident edges produces a subgraph with more connected components (i.e. a subgraph that is not connected).

- Such vertices are **cut vertices** (or **articulation points**). Example: Router
- Such edge is called a **cut edge** or **bridge**. Example: Link

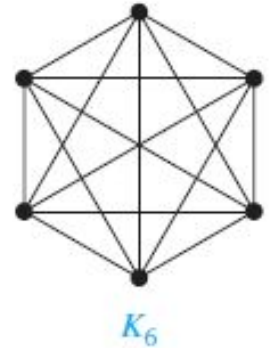
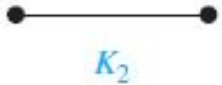
Cut Vertices and Cut Edges

Example: Find the cut vertices and cut edges in the graph G_1



Complete Graph

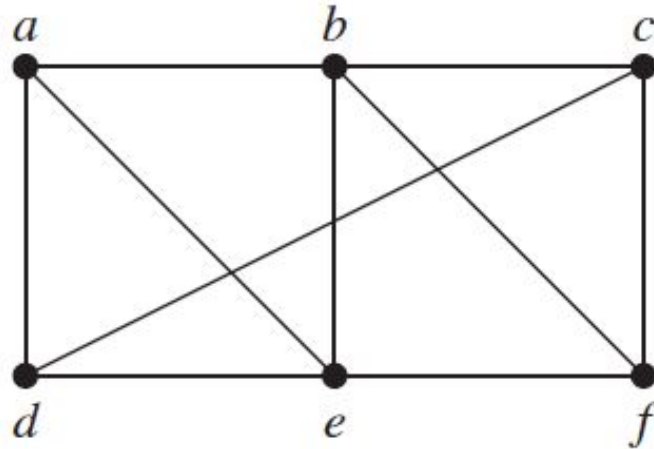
For the complete graph K_n , where $n \geq 3$, removing a vertex from K_n and all edges incident to it, the resulting subgraph is the complete graph K_{n-1} , a connected graph.



Vertex Cut

Definition 6: A subset V' of the vertex set V of $G = (V, E)$ is a **vertex cut**, or **separating set**, if $G - V'$ is disconnected.

- For instance, the set $\{b, c, e\}$ is a vertex cut with 3 vertices.



Vertex Connectivity

Definition 7: We define the **vertex connectivity** of a noncomplete graph G , denoted by $\kappa(G)$, as the minimum number of vertices in a vertex cut.

Example: When G is a complete graph, removing any subset of its vertices and all incident edges still leaves a complete graph.

- Instead, we set $\kappa(K_n) = n - 1$, the number of vertices needed to be removed to produce a graph with a single vertex.

Vertex Connectivity

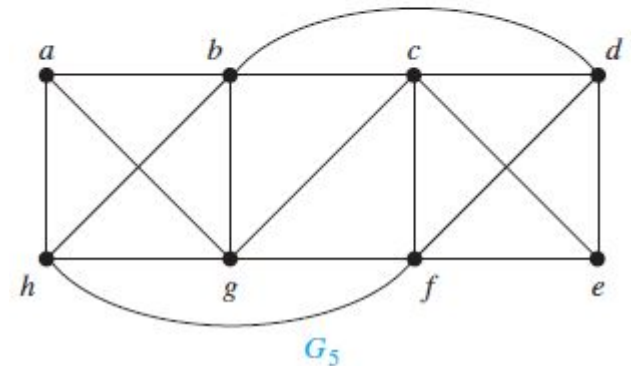
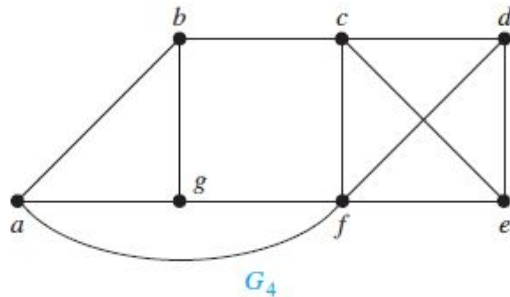
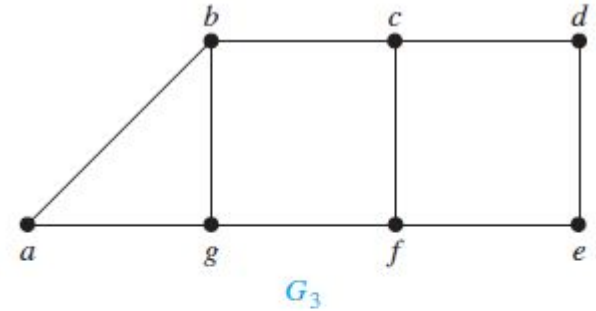
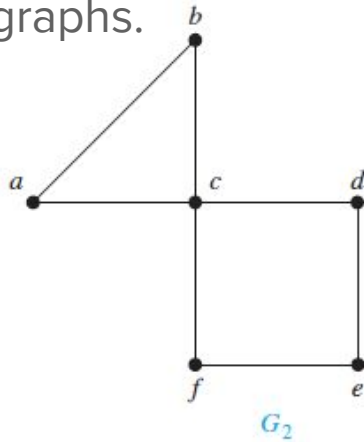
For every graph G , $\kappa(G)$ is minimum number of vertices that can be removed from G to either disconnect G or produce a graph with a single vertex.

- If G has n vertices, $0 \leq \kappa(G) \leq n - 1$.
- $\kappa(G) = 0$ if and only if G is disconnected.

The larger $\kappa(G)$ is, the more connected G to be.

Vertex Connectivity Example

Example: Find the cut vertices or vertex cut, and vertex connectivity $\kappa(G)$ for each of the graphs.



Edge Connectivity

Definition 8: We can also measure the connectivity of a connected graph $G = (V, E)$ in terms of the minimum number of edges that we can remove to disconnect it.

If a graph has a **cut edge**, then we need only remove it to disconnect G .

If G does not have a cut edge, we look for the smallest set of edges that can be removed to disconnect it.

A set of edges E' is called an **edge cut** of G if the subgraph $G - E'$ is disconnected.

Edge Connectivity

The **edge connectivity** of a graph G , denoted by $\lambda(G)$, is the minimum number of edges in an edge cut of G .

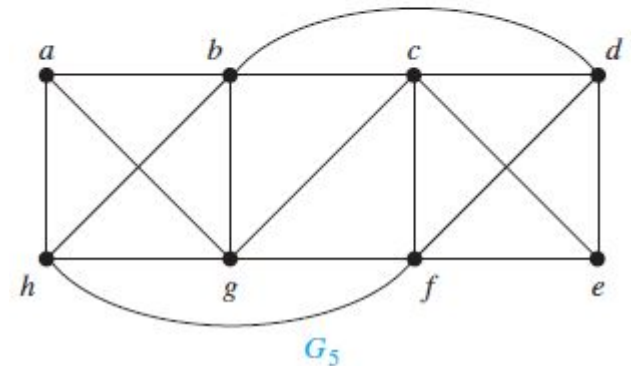
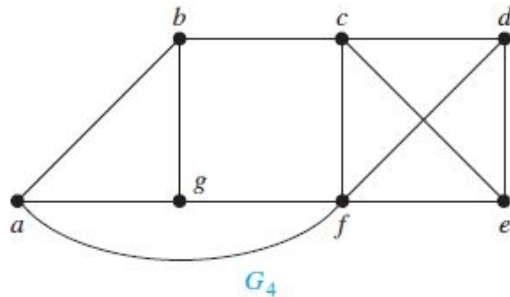
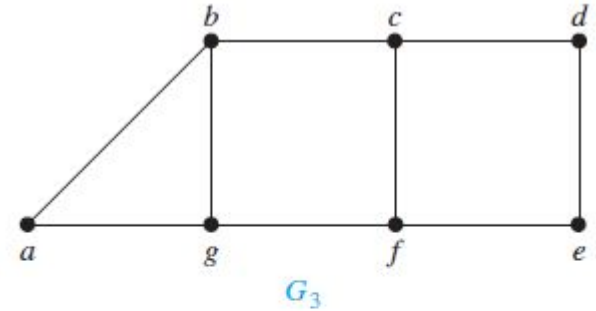
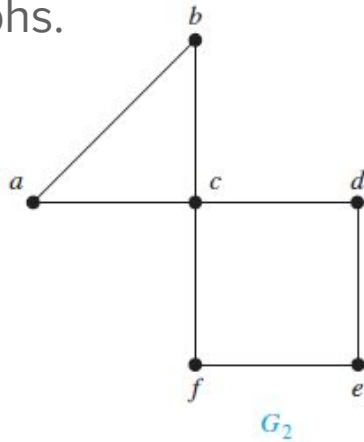
$\lambda(G) = 0$ if G is not connected.

Also, $\lambda(G) = 0$ if G is a graph consisting of a single vertex.

- If G is a graph with n vertices, then $0 \leq \lambda(G) \leq n - 1$.

Edge Connectivity Example

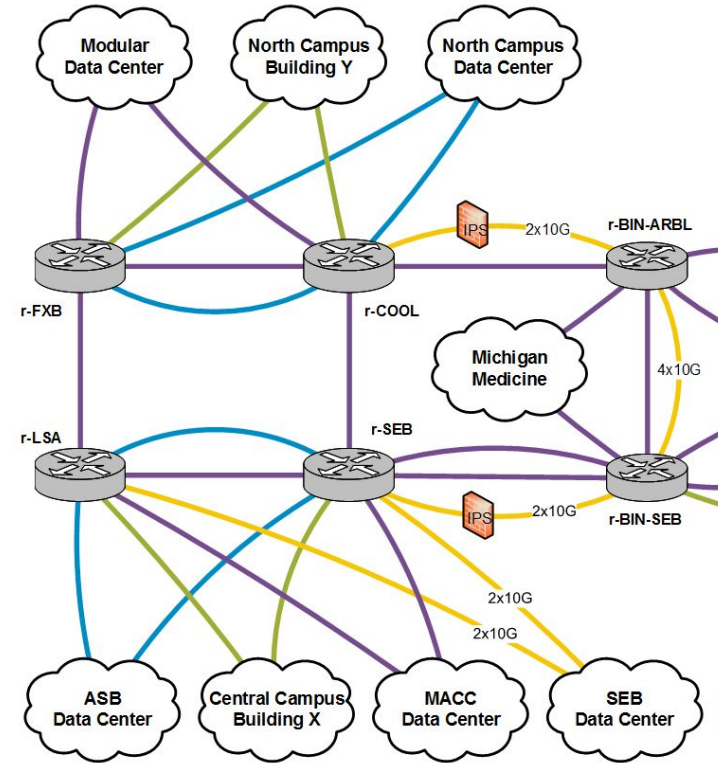
Example: Find the cut edges or edge cut, and edge connectivity $\lambda(G)$ for each of the graphs.



Applications of Vertex and Edge Connectivity

Graph connectivity plays an important role in reliability of networks.

- A data network can be modeled using vertices to represent **routers** and edges to represent **links** between them.



Ref: <https://its.umich.edu/enterprise/wifi-networks/campus-network-diagram-description>

Dr. Sirasit Lochanachit

Applications of Vertex and Edge Connectivity

A road network can be modeled using vertices to represent **intersections** and edges to represent **roads** running between intersections.



Ref: <https://twitter.com/doverview/status/590242249201602560>